CAPSTONE EXECUTIVE BRIEFING • FINAL REVIEW

Uber Eats Operational & Customer Intelligence System

An End-to-End Machine Learning Platform: Behavioral Segmentation, NLP Feedback Mining, Delivery Duration Regression, and Hourly Demand Forecasting.

Presenter: Trainee Data Analyst

Timeline: 7–Day Capstone Delivery

1. STRATEGIC CONTEXT

Core Marketplace Friction Points

Balancing customer loyalty, kitchen prep, and transit logistics

Customer Churn

Unmonitored drop-off across inactive customer cohorts.

Delivery Inaccuracy

Static delivery ETAs creating cold-food complaints.

Supply Imbalances

Unprepared courier fleets during peak dinner hours.

2. RELATIONAL FOUNDATION

Platform Data Schema

Synthetic generation modeled after real-world marketplace distributions

5,000	500	50	4,766
Total Orders	Active Customers	Restaurants	Delivered Trips

Primary Entity	Core Predictors & Signals	Business Usage
customers.csv	Age, City Zone, Account Age	Demographic Tiering
deliveries.csv	Distance, Prep Time, Weather, Traffic	ETA Duration Modeling
reviews.csv	1–5 Star Ratings, Text Feedback	Customer Sentiment Mining

3. PREPROCESSING RIGOR

Data Hygiene & Feature Pipelines

Transforming raw transactions into scaled analytical feature sets

Hygiene Protocol

Missing Values: Imputed review scores with median.

Standardization: Scaled features for distance models.

Identifier Stripping: Removed surrogate IDs before training.

Engineered Predictors

Behavioral Cadence: Days between repeat order transactions.

Temporal Flags: Peak lunch and dinner hour indicators.

Environmental Penalties: Rain and traffic delay factors.

4. CUSTOMER INTELLIGENCE

Customer Behavioral Segmentation

Discovered via K-Means (k=3), DBSCAN outlier filtering, and PCA

CLUSTER 1 (27.8%)

Premium Diners

13.8 orders, $554 lifetime spend.

CLUSTER 2 (46.0%)

Steady Regulars

9.7 orders, $373 lifetime spend.

CLUSTER 0 (26.2%)

Occasional Diners

6.5 orders, 22-day reorder gap.

DBSCAN Anomaly Detection: 25 Outlier Accounts Isolated

Identified bulk corporate buyers and extreme delivery dissatisfaction cases.

5. NLP FEEDBACK INTELLIGENCE

Sentiment Modeling & Failure Root Causes

Text classification on 2,860 customer reviews via TF-IDF & Linear SVM

Classifier Scorecard

98.3%

Linear SVM Test Accuracy

Macro F1-Score: 0.98

Negative Recall: 1.00 (Zero missed churn signals)

Primary Complaint Drivers



6. SUPERVISED REGRESSION

Doorstep Delivery Time Prediction

Benchmarking 4 supervised regression algorithms on 4,766 completed deliveries

Model Architecture	MAE	RMSE	R^2 Score	Production Status
Ridge Regression (L2)	**2.32 min**	**2.93 min**	**0.938**	**Selected Best Model**
Baseline Linear (OLS)	2.32 min	2.93 min	0.938	Collinear Sensitivity
Lasso Regression (L1)	2.54 min	3.19 min	0.926	Excessive Shrinkage
Random Forest (100 Trees)	2.80 min	3.55 min	0.909	High Latency Overkill

Dominant Predictors: Distance (58.6%), Kitchen Prep (17.7%), Traffic (+15.5 min), Rain (+7.0 min).

Tip Correlation: 63.4% with Basket Size

7. TIME-SERIES FORECASTING

Hourly Demand Trajectory

ARIMA(2, 0, 1) and Meta Prophet evaluated across 4,312 continuous hours

p < 0.001	1.09	42%	75.6%
ADF Test: Stationary (d=0)	ARIMA MAE (Orders/Hour)	Weekend Dinner Share	Downtown & West End Volume

Lunch Rush (12:00 – 14:00)

Demand jumps 3.5× over morning baseline.

Dinner Peak (18:00 – 21:00)

Heaviest delivery queues across Friday through Sunday.

END-TO-END ORCHESTRATION

Production Pipeline Flow

Automated multi-task architecture executed in src/pipeline.py

STAGE 1

Data Ingestion

Validation of 7 relational tables.

STAGE 2

Feature Engine

Scaling, dummies, TF-IDF vectors.

STAGE 3

Model Engine

Ridge ETA, tips, review sentiment.

STAGE 4

Dispatch Action

Dynamic rain buffers & fleet staging.

8. STRATEGIC RECOMMENDATIONS

Operationalizing Model Predictions

Direct translation of analytical outputs into platform profit and retention

1. Weather-Adaptive ETA Buffers

Add +7 min rain buffers to eliminate cold-food churn.

2. Synchronized Driver Dispatch

Dispatch couriers when kitchen prep is 75% complete.

3. Dynamic Fleet Staging

Stage 65% of courier fleet in Downtown by 5:30 PM.

4. Merchant Quality Interventions

Enforce 2-step verification for sides on 24 outlier merchants.

9 & 10. GOVERNANCE & ROADMAP

Model Limitations & Future Architecture

Risk boundaries today and deep learning expansion tomorrow

⚠ Known System Limitations

Synthetic Uniformity: Lacks multi-floor apartment delivery friction.

Cold-Start Gate: Requires 3+ transactions to tier new customers.

Lexical Rigidity: TF-IDF cannot detect nuanced sarcasm.

Future Technical Scope

DistilBERT Embeddings: Context-aware customer review sentiment.

Graph Neural Networks: Turn-by-turn road network routing.

Exogenous Regressors: Live weather and concert event feeds.

CAPSTONE CONCLUDED

Questions & Discussion

Uber Eats Operational & Customer Intelligence System

github.com/srishtijalan-dev/uber-eats